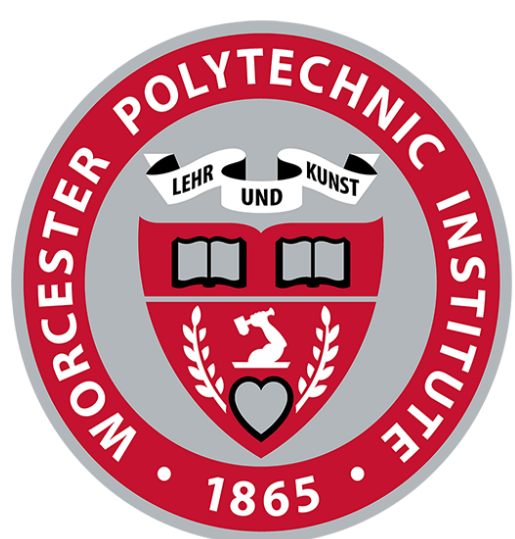


PRISM: An Open-Source Reference Chip for Reproducible Side-Channel Assessment of Post-Quantum Standards

Dillibabu Shanmugam, Patrick Schaumont

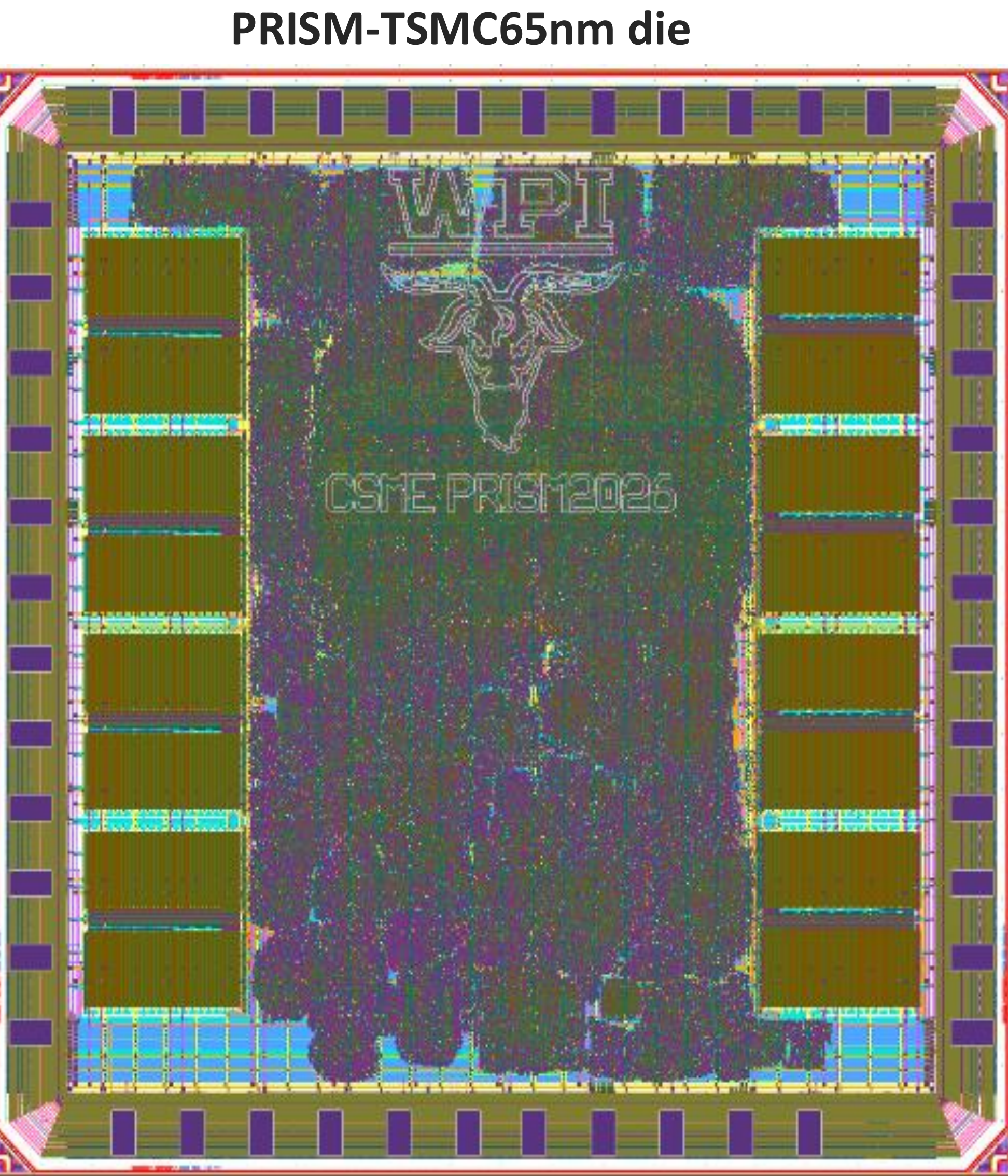
{dshanmugam, pschaumont}@wpi.edu



WPI

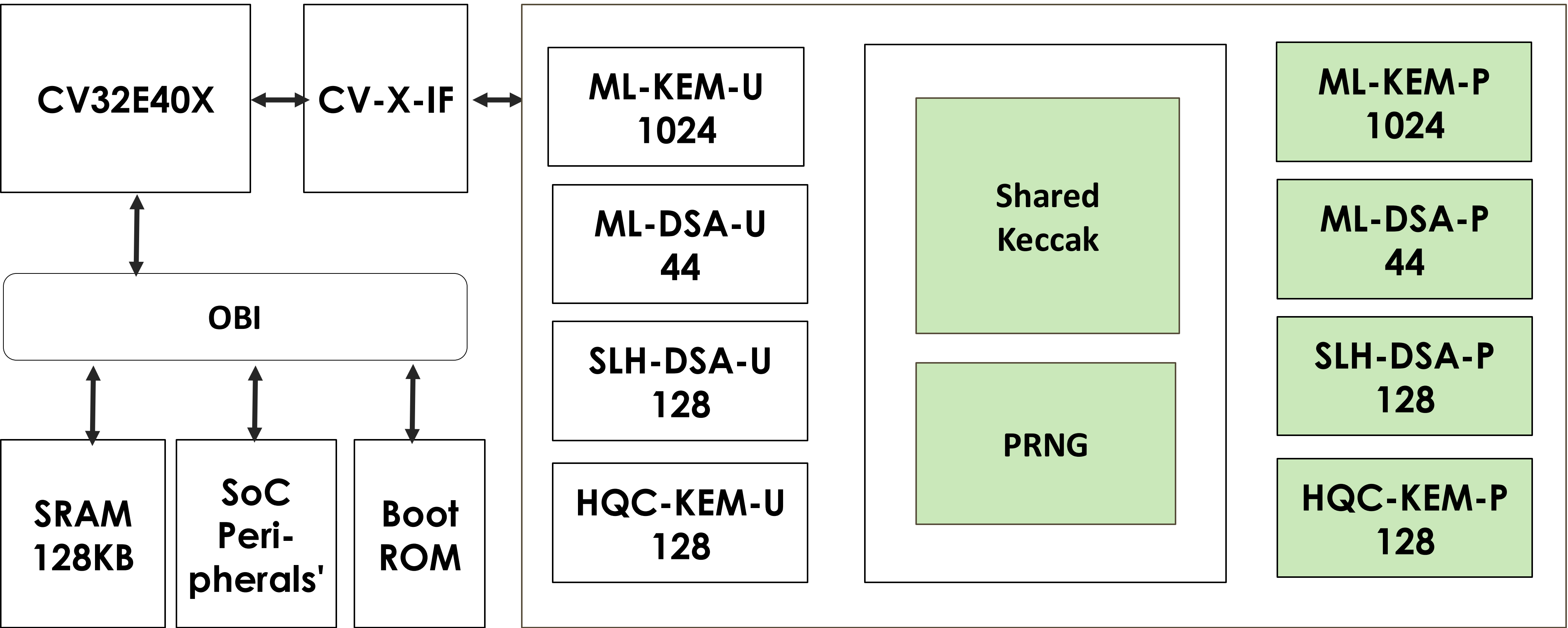
PRISM

PRISM is a unified open-source reference of protected and unprotected PQC implementations. PRISM supports the development of standard evaluations for PQC implementation attacks including fault injection and side-channel analysis



- FIPS 203, 204, 205, HQC : unprotected and protected variants
- Pre- and post-silicon SCA and fault platform
- Observability: scan chain over all 30301 flops
- Share-0 and share-1 registers placed apart to reduce coupling
- Unprotected ISE 3 to 41x faster than software on the same core

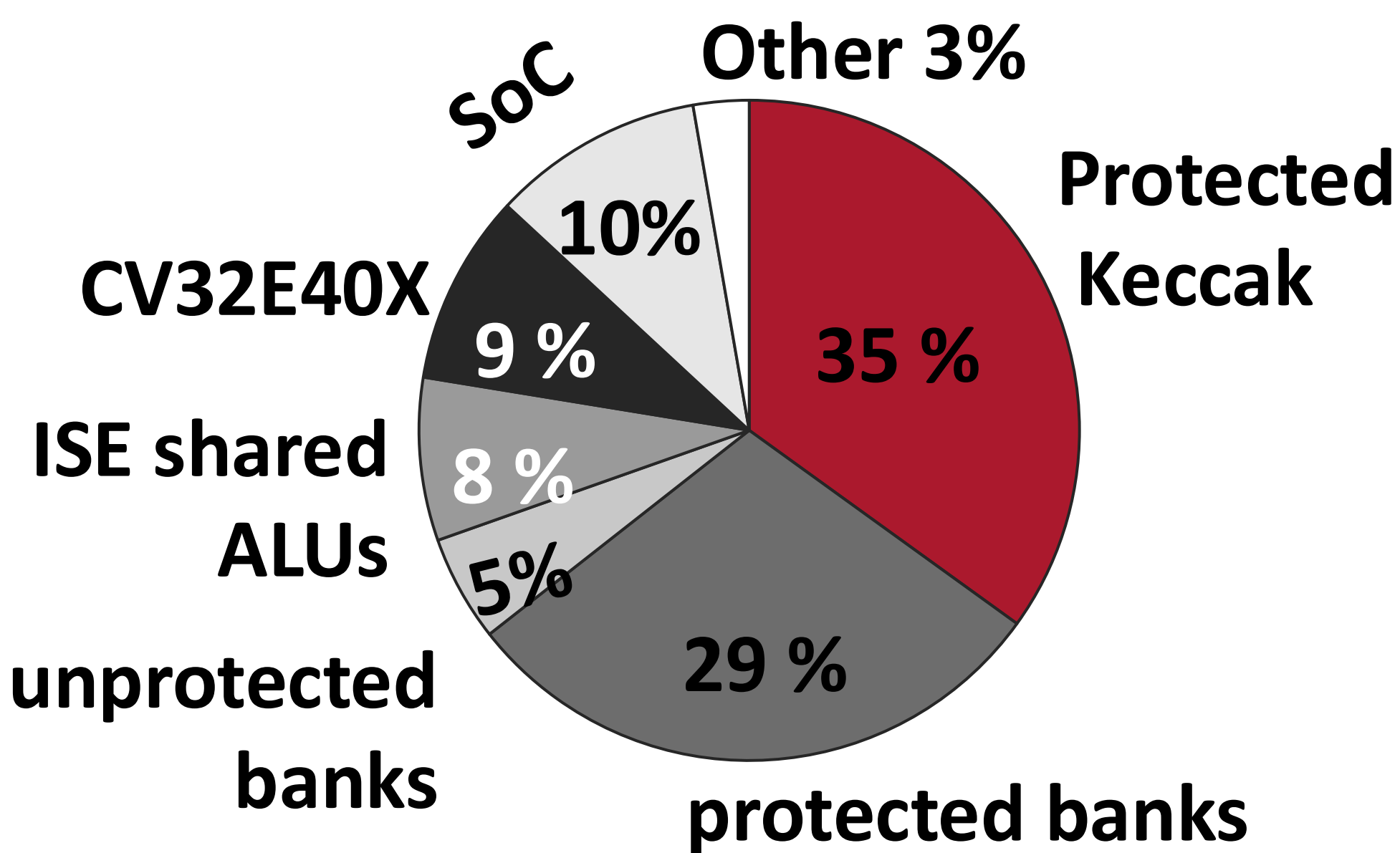
Architecture



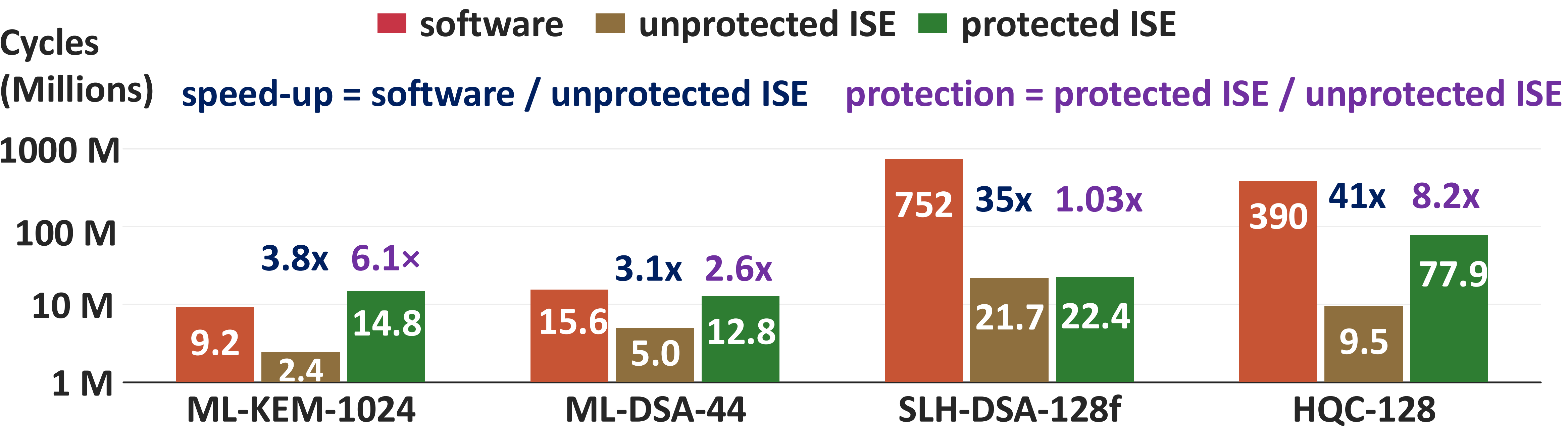
Design Decisions

- Protected and unprotected, same die
- Shared ALU, private masking gadgets
- One protected Keccak for all banks
- 2 to 135 cycles per ISE instruction
- Host-seeded PRNG with mask-off

Logic area : 152K cells



Performance



Comparison

	HORCRUX	OpenTitan	PRISM
Algorithms	ML-KEM, ML-DSA, SLH-DSA, HQC	ML-KEM, ML-DSA, SLH-DSA(SW)	ML-KEM, ML-DSA, SLH-DSA, HQC-KEM
Countermeasure	None	Protected: MLKEM, MLDSA	Protected : ML-KEM, ML-DSA, SLH-DSA, HQC
ISE	32	Coprocessor firmware	85
Clock frequency	160 MHz : Synthesis	100 MHz	48 MHz
Speed-up	2 to 3x	Not reported	3 to 41x
Chip	Not fabricated	Earl Grey	PRISM
Pre-silicon SCA	No	formal masking checks	Simulation

Future work : Evaluate the protected coprocessors against side-channel attacks on the PRISM-65 chip.

Tape-out 9/2/26, Chips Expected 12/2/26 <https://github.com/Secure-Embedded-Systems/ospqc-pro>